

Groups, graphs, and morphisms

Zayn Patel
zaynpatelwhs@gmail.com

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Cayley Digraphs begin with a group and represent it as a graph but we can also begin with a graph (non-Cayley) and relate it to a group. We will learn about the automorphism group of a graph and find isomorphisms between a graph automorphism and a well-known group.

Topics covered

To accomplish our goal, we will cover the following:

- ▶ Generators and relations
- ▶ Cayley Digraphs of Groups
- ▶ Hamiltonicity of Digraphs
- ▶ Graph theory meets groups theory
- ▶ Generators and relations rigor

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- ▶ Groups with a single element that combines with itself to generate the entire group are called **cyclic**. Recall that we can also have subgroups which are cyclic.
- ▶ Most groups we have worked with are not cyclic and have multiple generators. D_4 has R_{90} and H as generators.
- ▶ We call the set of generators of G the **generating set** which is usually denoted S . For D_4 , $S = \{R_{90}, H\}$.

Generators and relations

- Relations definition: the smallest set of equations that specify the conditions the generators must satisfy. In D_4
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- ▶ If two groups are defined by the same relations, they are isomorphic.
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- ▶ Describing a group by its generators and relations is called a **group presentation**. The presentation for D_4 is $\langle R, H | R^4 = H^2 = (RH)^2 = R_0 \rangle$.
- ▶ In order to define generators and relations rigorously, we need a beautiful concept called free groups. More information is available in the appendix if you are interested.

Cayley Digraphs of Groups

Purpose: Visualize a group given by generators and relations
These are directed graphs where vertices are group elements and arcs, which are the generators, connect the vertices.

Cayley Digraphs of Groups

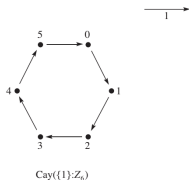
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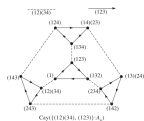
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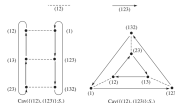
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(a) $\mathbb{Z}_6$ Digraph



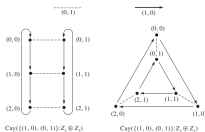
(b) A_4 Digraph



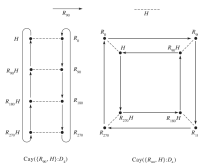
(c) S_3 Digraph

Cayley Digraphs of Groups

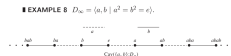
Example 8 shows us that we can make Digraphs for infinite groups too.



(a) $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ Digraph



(b) D_4 Digraph



(c) Infinite Digraph

Cayley Digraphs of Groups

Definition: Cayley Digraph of a Group

Let G be a finite group and let S be a set of generators for G . We define a Digraph $\text{Cay}(S : G)$, called the Cayley Digraph of G with the generating set S , as follows:

1. Each element of G is a vertex of $\text{Cay}(S : G)$.
2. For each x and y in G , there is an arc from x to y if and only if $xs = y$ for some $s \in S$.

Cayley Digraphs of Groups

Example

Let us look at $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ as an example. We need to check the first condition: each element of G is a vertex of $\text{Cay}(S : G)$.

$$\mathbb{Z}_3 \oplus \mathbb{Z}_2 = \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1)\}$$

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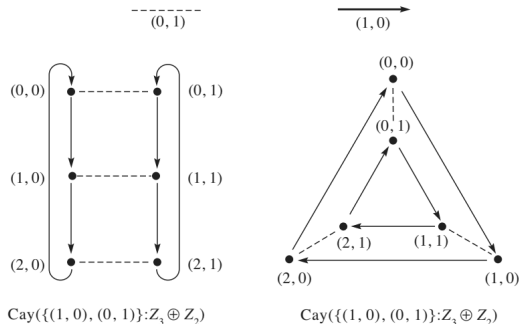


Figure: $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ Digraph

Cayley Digraphs of Groups

The second condition says that $\exists$ arc iff an element of the group multiplied by some generator produces a new element of the group.

Cayley Digraphs of Groups

The second condition says that $\exists$ arc iff an element of the group multiplied by some generator produces a new element of the group. Some examples of this in $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ are:

$$(2, 0) \cdot (1, 0) = (2 + 1, 0 + 0) = (3 \bmod(3), 0 \bmod(2)) = (0, 0)$$

$$(1, 1) \cdot (0, 1) = (1 + 0, 1 + 1) = (1 \bmod(3), 2 \bmod(2)) = (1, 0)$$

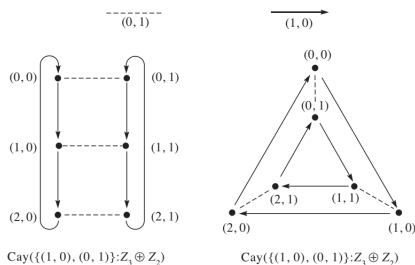


Figure: $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ Digraph

Cayley Digraphs of Groups

How do we find the relations for a group, and how might a graph help us do this?

Cayley Digraphs of Groups

How do we find the relations for a group, and how might a graph help us do this?

In general, finding the relations for a group is a problem in computational group theory. There are specific algorithms that search the "group space" to find and confirm relations in large groups. We will not cover these today. Instead, we will look at a finite group whose relations we can see by tracing the arcs.

Cayley Digraphs of Groups

We will use A_4 as an example and let us begin by finding the relations between the generators and the identity element.

Let $a = (12)(34)$ and $b = (123)$.

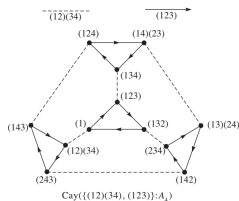


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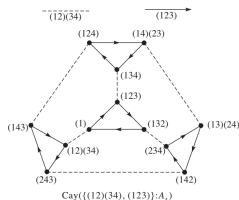


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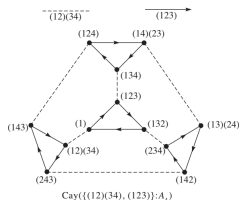


Figure: A_4 Digraph

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We can also derive other relations from this set of generators.

$$(ab)^3 = ababab = bab = a^{-1}b^{-1}a^{-1} = (124)$$

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Gallian mentions that there are cases where one group can be defined with two different sets of generators and relations. So another undecidable problem is determining whether these groups are isomorphic to each other.

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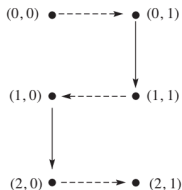
Lastly, note that defining a group by its presentation does not provide information on the cardinality of the group. Consider the group presentation for A_4 : $\langle a, b | a^2 = b^3 = (ab)^3 = e \rangle$. If we did not know prior information about the cardinality of A_4 we would not know if it was finite or infinite.

$\mathbb{Z}$ has the group presentation $\langle x \rangle$ and we know this is infinite.

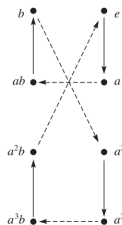
Graph Theory meets Group Theory

Definition: Hamiltonian Circuit and Path

- *Hamiltonian Circuit*: Path that starts at a vertex, visits each vertex exactly once, and returns to the starting vertex.



(a) Hamiltonian Path
in the $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ Digraph

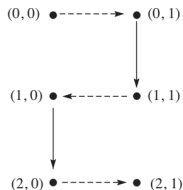


(b) Hamiltonian Circuit
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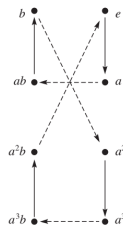
Graph Theory meets Group Theory

Definition: Hamiltonian Circuit and Path

- ▶ *Hamiltonian Circuit*: Path that starts at a vertex, visits each vertex exactly once, and returns to the starting vertex.
- ▶ *Hamiltonian Path*: Path that starts at a vertex and visits each vertex exactly once without returning to the starting vertex.



(a) Hamiltonian Path
in the $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ Digraph



(b) Hamiltonian Circuit
in the Q_4 Digraph

Graph Theory meets Group Theory

Hamiltonian paths and circuits are not unique; in any given graph, we may find multiple paths that satisfy the definition.

There is a condition about Hamiltonian circuits and generators that is important enough to prove.

Claim: If $s_1, s_2, \dots, s_n$ is a sequence of generators determining a Hamiltonian circuit, then the same sequence determines a Hamiltonian circuit for any starting vertex.

Graph Theory meets Group Theory

Proof.

Let $s_1, s_2, \dots, s_n$ be a sequence of generators determining a Hamiltonian circuit. In addition, let $v_1, v_2 \in G$. Since we have a Hamiltonian circuit we know that $v_1 s_1 s_2 \dots s_n = v_1 e$. By the cancellation property for groups $s_1 s_2 \dots s_n = e$.

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Now take another vertex, v_2 and let v_2 be the starting vertex. By the same argument as v_1 we know that $v_2 s_1 s_2 \dots s_n = v_2 e$. After applying the cancellation property $s_1 s_2 \dots s_n = e$.

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Thus we have shown that for two different, arbitrary starting vertices the same sequence results in a Hamiltonian circuit. □

Hamiltonian circuits in $\text{Cay}(\{(1, 0), (0, 1)\} : \mathbb{Z}_m \oplus \mathbb{Z}_n)$

In our examples we looked at $\mathbb{Z}_3 \oplus \mathbb{Z}_2$, let us investigate if there is a Hamiltonian circuit in its Cayley Digraph. We will first consider the case where $\gcd(m, n) = 1$.

Claim: There is no Hamiltonian circuit in $\mathbb{Z}_m \oplus \mathbb{Z}_n$ when $\gcd(m, n) = 1$.

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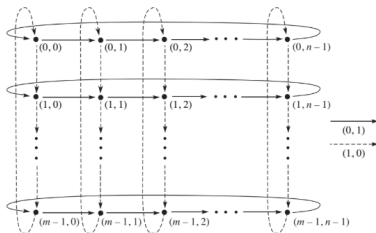


Figure 28.4 $\text{Cay}(\{(1, 0), (0, 1)\} : \mathbb{Z}_m \oplus \mathbb{Z}_n)$.

Figure: $\mathbb{Z}_m \oplus \mathbb{Z}_n$ Digraph as a rectangular grid

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Proof.

Suppose there is a Hamiltonian circuit in the Digraph and we start at some vertex (a, b) which exits horizontally.

If (a, b) exits horizontally then $(a - 1, b + 1)$ must also exit horizontally otherwise it will pass through $(a, b + 1)$ twice. Now, $(a - 2, b + 2)$ must exit horizontally otherwise it will pass through $(a - 1, b + 1)$ twice.

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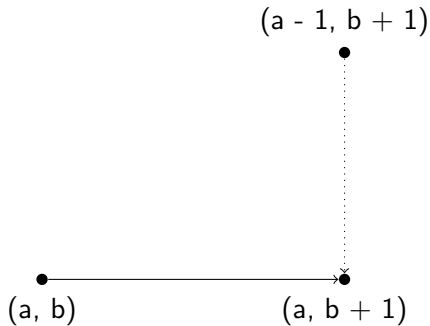
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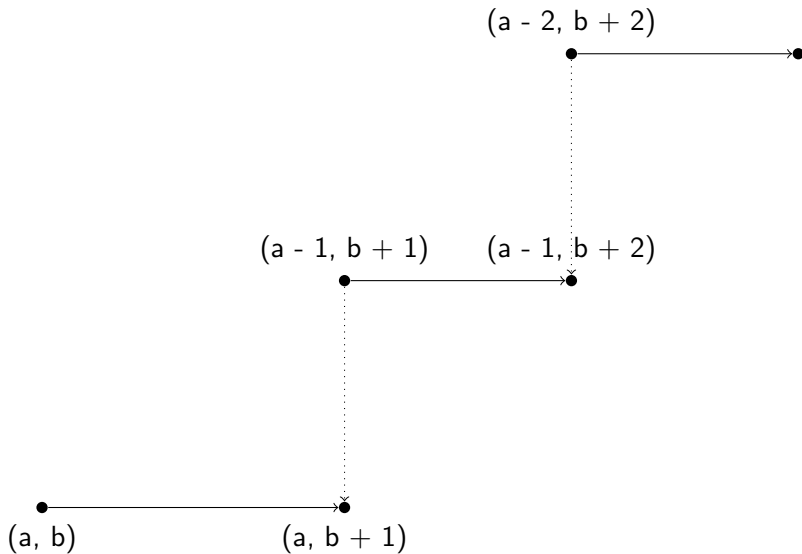
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When we repeat this argument we see that every vertex will need to exit horizontally. But, if we exit each vertex horizontally we cannot have a Hamiltonian circuit. Thus we get a contradiction; there is no Hamiltonian circuit for a Digraph with $\gcd(m, n) = 1$. $\square$

Visualizing our proof



Visualizing our proof



Hamiltonian circuits in $\text{Cay}(\{(1, 0), (0, 1)\} : \mathbb{Z}_m \oplus \mathbb{Z}_n)$

Now we look at the special case where $\gcd(m, n) > 1$.

Claim: $\text{Cay}(\{(1, 0), (0, 1)\} : \mathbb{Z}_m \oplus \mathbb{Z}_n)$ has a Hamiltonian circuit when $n \mid m$.

Proof.

By hypothesis we know that $m = nk$ for some $k \in \mathbb{Z}$. We want to prove the claim using a coordinatized Digraph similar to our previous theorem. Since the $\gcd(m, n) > 1$ we have an external direct product of the form $\mathbb{Z}_{nk} \oplus \mathbb{Z}_n$. This product has $n^2 k$ elements. The external direct product has $n^2 = n \times n$ elements k times. So the coordinatized Digraph has k $n \times n$ blocks.

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Proof.

To form a Hamiltonian circuit we begin at the vertex $(0, 0)$ and cover the vertices in the first block by using the generator $(0, 1)$ until the last vertex in the first row is reached. Then apply the generator $(1, 0)$ to move vertically and continue this process of applying $(0, 1)$ to move horizontally and $(1, 0)$ to move vertically for each of the k blocks until the vertex $(n - 1, 0)$ has been reached. We move vertically to the next block and repeat this process until the last vertex in the k th $n \times n$ block.

Hamiltonian circuits in $\text{Cay}(\{(1, 0), (0, 1)\} : \mathbb{Z}_m \oplus \mathbb{Z}_n)$

Now we look at the special case where $\gcd(m, n) > 1$.

Claim: $\text{Cay}(\{(1, 0), (0, 1)\} : \mathbb{Z}_m \oplus \mathbb{Z}_n)$ has a Hamiltonian circuit when $n \mid m$.

Proof.

Now, the left-most vertex in the last block is $(nk - 1, 0)$ and this needs to exit horizontally or vertically, by definition. Apply the generator $(1, 0)$ to this vertex. Then

$(nk - 1, 0) + (1, 0) = (nk - 1 + 1, 0) = (nk, 0)$. Recall that our external direct product is $\mathbb{Z}_{nk} \oplus \mathbb{Z}_n$ so we can apply the group operation to nk s.t. $nk \bmod (nk) = 0$.

Thus, $(nk - 1, 0)$ exits vertically to $(0, 0)$ completing the Hamiltonian circuit. □

Visualizing our proof with $\mathbb{Z}_{3k} \oplus \mathbb{Z}_3$

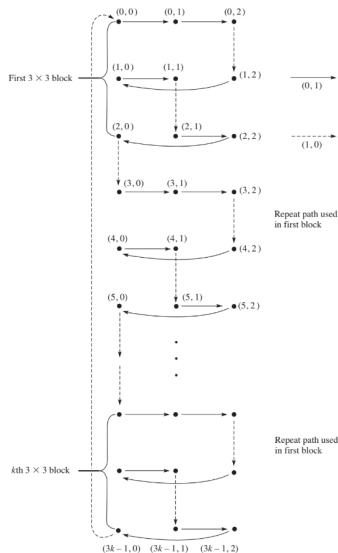


Figure: Coordinatized $\mathbb{Z}_{3k} \oplus \mathbb{Z}_3$ Digraph

Additional details about $\gcd(m, n)$

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- ▶ In the relatively prime graph each vertex was in the coset $(a, b) + \langle(-1, 1)\rangle$. This is because in the relatively prime case there was no generator that gave us a multiple of a and b at the same time so that we can reduce it to the starting vertex: (a, b) . *As an example, try adding either generator to the last element in the $\mathbb{Z}_3 \oplus \mathbb{Z}_2$ Digraph.*

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- ▶ The two theorems we proved are about external direct products of two $\mathbb{Z}$ -mod groups but I wonder if we can generalize this theorem to Cayley Digraphs with more than two groups.

Abelian Groups Have Hamiltonian Paths

Theorem

Let G be a finite Abelian group, and let S be any generating set for G . Then $\text{Cay}(S : G)$ has a Hamiltonian path.

Proof.

We use induction on $|S|$. If $|S| = 1$, say, $S = \{a\}$, then the Digraph is just a circle with labels $e, a, a^2, \dots, a^{m-1}$, where $|a| = m$. There is a Hamiltonian path for this case; it is just the path around the circle.

Now assume that $|S| > 1$. Choose some $s \in S$. Let $T = S - \{s\}$. Define $H = \langle s_1, s_2, \dots, s_{n-1} \rangle$ where $H \leq G$ and H could be equal to G . H contains all possible combinations of the generators $s_1, s_2, \dots, s_{n-1}$, and although H contains one less generator than S we can still generate G because it is possible that S contains redundant generators. There is no constraint in the hypothesis that S is the minimal set of generators.

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Proof.

Because $|T|$ is less than $|S|$ and H is a finite Abelian group, the inductive hypothesis guarantees that there is a Hamiltonian path $(a_1, a_2, \dots, a_k)$ in $\text{Cay}(T : H)$.

The a_i represent the generators in T and their products give elements in H (since T generates H). We use this notation because we want to add generality about how the generators are combined to give an element $s_1^{i_1} s_2^{i_2} \dots s_{n-1}^{i_{n-1}} \in H$.

We want to show that there is a Hamiltonian path in $\text{Cay}(S : G)$. We will take advantage of the fact that we have a subgroup H and we can take any element $g \in G$ and form the left coset gH . Now note that gH is an element of G/H .

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Proof.

We can form this factor group because G is Abelian and its subgroups H are also Abelian and normal.

Each $g \in G$ looks like $g = s_1^{i_1} s_2^{i_2} \dots s_n^{i_n}$ since any element in G is a combination of elements in the generating set. So

$$gH = s_1^{i_1} s_2^{i_2} \dots s_n^{i_n} H = [(s_1^{i_1} s_2^{i_2} \dots s_{n-1}^{i_{n-1}})H] \cdot [s_n^{i_n} H].$$

But $(s_1^{i_1} s_2^{i_2} \dots s_{n-1}^{i_{n-1}}) \in H$ so this collapses to H or eH because of the coset property that $aH = H$ if $a \in H$.

Thus, we get $eH \cdot s_n^{i_n} H = s_n^{i_n} H$; this follows from coset multiplication rules. So the elements of G/H look like $s_n^{i_n} H$. The book notates this as $s^n H$ so we will use this notation for the remainder of the proof.

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Let G be a finite Abelian group, and let S be any generating set for G . Then $\text{Cay}(S : G)$ has a Hamiltonian path.

Proof.

Now we can see that $G/H = \{s^0H, s^1H, \dots, s^nH\}$. We conclude that sH generates the factor group G/H and also note that this partitions the group G into cosets of the form $s^0H, s^1H, \dots, s^nH$. We will use these cosets to form the Hamiltonian path in $\text{Cay}(S : G)$. We begin in the first coset, H , and start from the identity element of G , which is in H because $H \leq G$, the path given by $(a_1, a_2, \dots, a_k)$ visits each element of H exactly once because $(a_1, a_2, \dots, a_k)$ is a Hamiltonian path in $\text{Cay}(T : H)$. Because we partitioned G into these n cosets we are treating each distinct coset as a $\text{Cay}(T : H)$ Digraph.

Abelian Groups Have Hamiltonian Paths

Theorem

Let G be a finite Abelian group, and let S be any generating set for G . Then $\text{Cay}(S : G)$ has a Hamiltonian path.

Proof.

When we do this, the inductive hypothesis says that there exists a Hamiltonian path in $\text{Cay}(T : H)$, the generator S moves to the next coset and we apply the same idea until the last element in the last coset.

The vertices of $\text{Cay}(S : G)$ are elements of G and these elements are in exactly one coset sH^i because of the partition fact referenced earlier. Since we visited each element in each coset sH^i this implies that we visited each vertex of $\text{Cay}(S : G)$ exactly once. Thus, we have a Hamiltonian path. □

Visualizing our proof

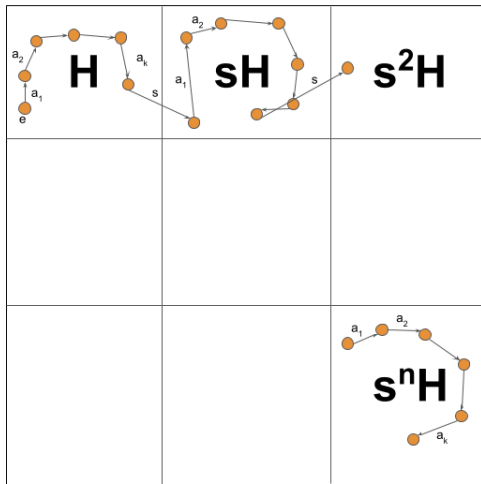


Figure: Coset Partition of $\text{Cay}(S : G)$

More group theory and graph theory connections

So far we have associated a graph with a group but we can go the other way. That is, we can associate a group with every graph. This group is called the automorphism group of a graph.

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We will be talking about the automorphism group of a graph, so we need to define new terms. From group theory, we know that an automorphism is an isomorphism from a group to itself. The set of automorphisms of a group forms a group called the **automorphism group** denoted $Aut(G)$. The elements of $Aut(G)$ are group isomorphisms.

The appendix contains a proof that the automorphism group is a group under function composition.

Graph Isomorphisms and Automorphisms

Definition: Graph Isomorphisms and Automorphisms

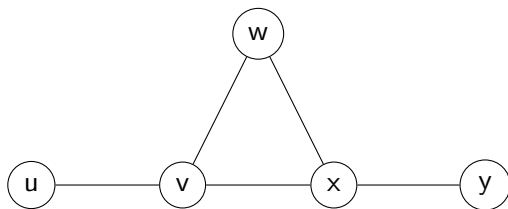
- *Graph isomorphism:* An isomorphism between graphs G and H is a bijection $\phi : V(G) \rightarrow V(H)$ s.t. $uv \in E(G)$ iff $\phi(u)\phi(v) \in E(H)$. Notice that this defines an isomorphism between the vertex sets of the graphs. In examples, we will see that the degree of the vertex also matters in defining an isomorphism.

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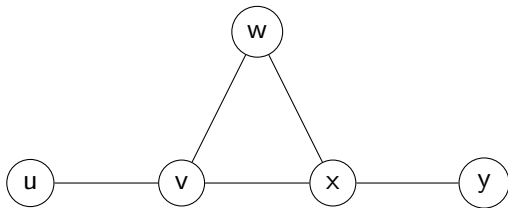
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- ▶ *Graph automorphism:* An isomorphism from a graph to itself. Formally, it is a bijection $\phi : V(G) \rightarrow V(G)$ s.t. $uv \in E(G)$ iff $\phi(u)\phi(v) \in E(G)$.

Finding the automorphisms of a graph G



We will denote the identity automorphism as ϵ with the map $u \rightarrow u, v \rightarrow v, x \rightarrow x, y \rightarrow y$.

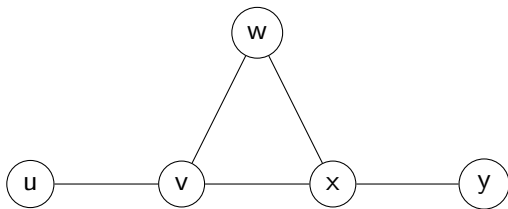
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α has the map $u \rightarrow y, y \rightarrow u, v \rightarrow x, x \rightarrow v, w \rightarrow w$. We map $u \rightarrow y$ because these are the only elements of degree one and $v \rightarrow x$ because these were adjacent to the elements of degree one. So, $\text{Aut}(G) = \{\epsilon, \alpha\}$.

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We can compose these using the group operation. But recall that $\text{Aut}(G)$ is a group and we know, from group theory, that the non-identity element is its own inverse in groups with two elements. Thus, $\alpha \circ \alpha = \epsilon$.

Finding the automorphisms of a graph G

Now that we know the elements of $Aut(G)$ and their products we can draw a Cayley table.

	ϵ	α
ϵ	ϵ	α
α	α	ϵ

What well known group with two elements equals the identity when we add the non-identity element to itself?

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Conclusion: $Aut(G) \cong \mathbb{Z}_2$. We have found an automorphism between a graph and a well-known group!

Equivalence relation between two vertices in G

Let $u, v \in V(G)$. We say $u \sim v$ if $\exists \phi \in \text{Aut}(G)$ s.t. $\phi(u) = v$.

Two vertices are related if there is some graph automorphism that maps one vertex to the other.

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Application: Efficient Graph Automorphism by Vertex Partitioning

The graph automorphism partition has other uses. Since it is in essence an encoding of the symmetries of the graph, having the automorphism partition available can be of definite aid ...

Symmetry information contained in the automorphism partition can also be used to prune backtracking tree searches.

The tree search can maintain the automorphism partition of the data structures searched thus far. Then, when new structures are generated, the symmetries in the partition can be used to detect and eliminate redundant branches in the search.

We now know:

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- ▶ Definition and examples of a Cayley Digraph for finite and infinite groups
- ▶ Finite Abelian Groups have Hamiltonian Paths
- ▶ We can form a group (automorphism group) based on the vertex set of a graph. There may be cases where this group is isomorphic to more well known groups.

Acknowledgments

- ▶ Thank you Dr. Casey Mann for your time and willingness to dive into the details of Theorem 28.3, equivalence relations of words, and other topics in this presentation. I left all of our office hour sessions with more clarity on how to think about these results.
- ▶ Thank you Dr. Joseph A. Gallian for writing a sublime Abstract Algebra textbook and specifically for the exposition on generators and relations and Cayley Digraphs of Groups.
- ▶ Thank you to the authors of the many online sources I used and learned from while creating this talk. I have linked their work in the SOURCES.md file of this GitHub repository.

Free groups

The point of this section is to add a rigorous explanation to the generators and relations discussion that began the talk. To begin, we will define some terms about free groups, which is the set that contains generators and relations.

Let $S = \{a, b, c, \dots\}$ be a set of distinct symbols. As an example from permutation groups, S would not contain both $(12)(13) = (123)$ because these are obviously not distinct. This concept of distinctness is important for free groups.

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Define $S^{-1} = \{a^{-1}, b^{-1}, c^{-1}, \dots\}$ so that each element of S has an inverse in S^{-1} . We define the set $W(S)$ to be the collection of all finite strings of the form $x_1x_2\dots x_n$ where each $x_i \in S \cup S^{-1}$.

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The elements of $W(S)$ are called words from S . Notice that this set contains every possible combination of elements in S and S^{-1} .

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- ▶ Inverses are not as simple as aa^{-1} because a is just a symbol so aa^{-1} has no meaning.
- ▶ We need to define inverses so $W(S)$ is a group; we can do this with equivalence classes.

Proving an equivalence relation on $W(S)$

Definition: Equivalence class of words

For any pair of elements u and v of $W(S)$, we say that u is related to v if v can be obtained from u by a finite sequence of insertions or deletions of words of the form xx^{-1} or $x^{-1}x$ where $x \in S$.

Example: $S = \{a, b, c\}$

$acc^{-1}b$ is equivalent to ab because we can obtain ab by a deletion of the form cc^{-1} where $c \in S$.

Proving an equivalence relation on $W(S)$

Claim: $u \sim v$ if v can be obtained from u by a finite sequence of insertions and deletions.

Proof.

To prove that $u \sim v$ is an equivalence relation we need to prove that it satisfies reflexivity, symmetry, and transitivity.

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Transitivity ($u \sim v, v \sim w \implies u \sim w$): If $u_1u_2\dots u_n \sim v_1v_2\dots v_n$ and $v_1v_2\dots v_n \sim w_1w_2\dots w_n$ then there exist two different finite sequences of insertions and deletions. Concatenate these two sequences together; this result is exactly the relation $u \sim w$. □

Equivalence classes of $W(S)$ form a group

Theorem

Let S be a set of distinct symbols. For any word u in $W(S)$, let $\bar{u}$ denote the set of all words in $W(S)$ equivalent to u (i.e., $\bar{u}$ is the equivalence class containing u). Then the set of all equivalence classes of elements of $W(S)$ is a group under the operation $\bar{u} \cdot \bar{v} = \overline{uv}$.

- ▶ The elements of this group are **sets**. These sets are equivalence classes that partition $W(S)$.

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- ▶ The elements of this group are **sets**. These sets are equivalence classes that partition $W(S)$.
- ▶ This group is called a free group on S because it contains no additional relations which do not follow from the group axioms. As an example, it does not contain something like $ab = ba$.

Universal Mapping Property

Theorem

Every group is a homomorphic image of a free group.

Proof.

Let G be a group, and let S be a set of generators for G . Now, let F be the free group on S .

Before we continue the proof, we develop some notation to distinguish between the words in $W(S)$ and elements of G . We will denote the word $x_1x_2\dots x_n$ in $W(S)$ by $(x_1x_2\dots x_n)_F$ and the product in G by $(x_1x_2\dots x_n)_G$. $\overline{x_1x_2\dots x_n}$ is the equivalence class in F containing the word $(x_1x_2\dots x_n)_F$.

The elements $\overline{x_1x_2\dots x_n}$ and $(x_1x_2\dots x_n)_G$ are different elements because the operations on F and G are different. We know that the operation of F is the concatenation of the elements of the equivalence classes; the operation of G is unknown.

Universal Mapping Property

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Every group is a homomorphic image of a free group.

Proof.

Consider the mapping from F to G given by:

$\phi(\overline{x_1x_2\dots x_n}) = (x_1x_2\dots x_n)_G$. This maps the product of elements which form an equivalence class in F to the product in G .

Note that ϕ is well-defined, because inserting or deleting expressions of the form xx^{-1} or $x^{-1}x$ in elements of $W(S)$ corresponds to inserting or deleting the identity in G . Recall that we said that the inverses of symbols in F do not have any formal meaning *but* a product $x^{-1}x$ does have meaning inside of a group; those elements combine to give the identity (by group axioms).

Universal Mapping Property

Theorem

Every group is a homomorphic image of a free group.

Proof.

We can check that ϕ is operation-preserving, observe that

$$\begin{aligned}\phi(\overline{x_1 x_2 \dots x_n} \overline{y_1 y_2 \dots y_n}) &= \phi(\overline{x_1 x_2 \dots x_n y_1 y_2 \dots y_n}) = \\ (\overline{x_1 x_2 \dots x_n y_1 y_2 \dots y_n})_G &= (\overline{x_1 x_2 \dots x_n})_G (\overline{y_1 y_2 \dots y_n})_G.\end{aligned}$$

We move from the second to the third equality using the map ϕ and we get the last equality by the closure of G . Lastly, ϕ is onto G because S generates G . □

Takeaway: We have defined a free group on a set of generators and we found a homomorphic map which maps *onto* the group G . This shows that we can "hit" every element in G without any relations.

Universal Factor Group Property

Takeaway 2: The Universal Mapping Property (UMP) provided an onto function from $F \rightarrow G$ but this map can be smaller than G (if G is not taken to be the generating set) or larger than G . It may be larger since F contains all combinations of symbols, and some of these symbols might be redundant based on the relations of G . If we want to reconstruct G from a free group, we need the following theorem.

Theorem

Every group is isomorphic to a factor group of a free group.

We will not prove this; let us instead understand the theorem.

Understand the UFG Property

Recall the First Isomorphism Theorem for groups: Let ϕ be a group homomorphism from G to $\bar{G}$. Then the mapping from $G/\text{Ker}\phi$, given by $g\text{Ker} \rightarrow \phi(g)$, is an isomorphism. In symbols $G/\text{Ker}\phi \cong \phi(G)$.

Universal Factor Group Property

Understand the UFG Property

We want a homomorphism so we can use the First Isomorphism Theorem for groups. To get a homomorphism use the identity map $f : G \rightarrow G$. Every map of sets $f : G \rightarrow G$ extends to a unique homomorphism $\phi : F \rightarrow G$ if the generators of the domain are sent to the generators of the codomain. But this is obviously the case in f since each element of G is sent to itself.

Now we have a homomorphism ϕ . The definition of $\text{Ker}\phi = \{f \in F \mid \phi(f) = e_G\}$ says that we take any element of the free group and the homomorphism sends this element to the group identity.

Universal Factor Group Property

Understand the UFG Property

We saw something like this earlier. When we first began talking about relations these were defined as $\langle element = element = \dots = e \rangle$. From the takeaway of UMP we know that F contains all combinations of elements of G with no additional relations. But the $Ker\phi$ is naturally the set of relations for G since it sends these combinations from F to the groups' identity.

Thus, $F/Ker\phi \cong G$.

Generators and Relations Definition

We now have enough understanding to define generators and relations.

Generators and Relations

Let G be a group generated by some subset $A = \{a_1, a_2, \dots, a_n\}$ and let F be the free group on A . Let $W = w_1, w_2, \dots, w_t$ be a subset of F and let N be the smallest normal subgroup of F containing W . We say that G is given by the generators $a_1, a_2, \dots, a_n$ and the relations $w_1, w_2, \dots, w_t = e$ if there is an isomorphism from F/N onto G that carries $a_i N$ to a_i .

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- ▶ We define the free group on a set of distinct generators A .
- ▶ Relations are in a normal subgroup ($\text{Ker } \phi$ is normal so this aligns with First Isomorphism Theorem intuition).
- ▶ We needed the quotient group corollary to define an isomorphism from the factor group with $F \rightarrow G$.

Appendix: Proof that the automorphism group is a group

Claim: The automorphism group is a group.

Proof.

Let $\text{Aut}(G)$ be the automorphism group of G . We want to show that this is a group. First, we note that the identity map (the map that takes each element of G in the domain to its corresponding g in the codomain) is an element of $\text{Aut}(G)$ by definition. This map is 1-1, onto, and operation preserving. Thus, we have an identity element of $\text{Aut}(G)$.

Next, let $\varphi : G \rightarrow G, \lambda : G \rightarrow G \in \text{Aut}(G)$. We want to show that $(\varphi \circ \lambda)(g) \in G$. $\lambda(g) = g$ in $\lambda : G \rightarrow G$. So g is an element of the codomain of λ that is the input to φ . Then $\varphi(g) = g$ by its map. Thus, we have closure.

For inverses, let $\varphi : G \rightarrow G \in \text{Aut}(G)$. By the definition of $\text{Aut}(G)$ φ is an isomorphism. By Theorem 6.2, property 1 (Gallian 10th edition) φ^{-1} exists and is an isomorphism from $G \rightarrow G$. Lastly, the associative property is inherited from the definition of function composition. □